Abdallah Abdallah

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Summary

Final-year Computer Science and Engineering student (CGPA 3.60) with research experience on the reliability of AI agents and on grounding LLMs with symbolic structure, and a co-authored paper accepted at NILES 2026. Also built machine learning systems in industry at HERE Technologies, working with geospatial and vehicle-probe data. Interested in how AI agents behave and fail on real software tasks.

Education & Achievements

B.Sc. Computer Science and Engineering, Egypt-Japan University (E-JUST) Oct 2022 – Feb 2027 (exp.), CGPA 3.60
First Prize, HERE Cairo Geospatial Student Hackathon 2026.
Represented Egypt, HERE GeoAI Global Student Hackathon 2026.

Publications

S. M. Ammar, **A. A. Abdallah**, A. H. Abdelgawwad, M. M. Abdelbaset, M. A. Yehia, A. S. Farhan, A. M. Riyad, E. H. El-Shazly. *Revisiting Knowledge Distillation for Medical Vision–Language Retrieval: The Role of Contrastive Learning and Hard Negatives.* NILES 2026 (accepted; to appear in IEEE Xplore).

- Showed that feature imitation alone fails to preserve image–report retrieval ranking; a contrastive term plus semantic-aware hard-negative fine-tuning let lightweight students outperform the BioViL-T teacher on a 14,018-study benchmark, with a 5.1× smaller vision encoder.

Research Experience

Research Intern – University of Illinois Urbana-Champaign (UIUC++ SRSE Program) May 2026 – Aug 2026

- Worked in Prof. Tianyin Xu's group on the reliability and safety of AI agents for Site Reliability Engineering. Selected through a technical evaluation task, for which I designed and implemented a new SREGym problem (PR #835): a PodDisruptionBudget that silently deadlocks node drains while the application stays healthy, with a mitigation oracle that rejects partial fixes.
- Analyzed the Reddit Pi-Day outage and wrote a failure analysis plus an SREGym feasibility study that rates each causal step as exact, native, faithful proxy, or not reproducible, checked against the SREGym codebase.
- Built an MCP-based integration enabling the OpenSRE agent to investigate and act on live SREGym clusters, and added an agent-loop modification that makes it execute its own fixes rather than only recommending them (its loop otherwise stops once it can explain a fault). Demonstrated on two SREGym problems (missing ConfigMap, wrong service selector).

Undergraduate Thesis – E-JUST, advised by Prof. Ehab El-Shazly 2026 – present

- Lightweight vision-language models (Thesis I–II), extending the NILES 2026 work toward a planned journal paper.

Research Intern – Nile University Jul 2026 – Aug 2026

- Built a bilingual (Arabic/English) indoor-outdoor campus navigation system that grounds LLM reasoning in a Spatial Knowledge Graph, reducing route hallucination from 100% (ungrounded baselines) to 0%.
- Developed automated ETL pipelines to ingest and prune vector data from OpenStreetMap (OSM) and ArcGIS (GDB), constructing a 3D navigable topology with over 8,000 edges.

Work Experience

Data Science Intern – HERE Technologies Jul 2026 – Dec 2026 (expected)

- Built a road-centerline geometry pipeline from vehicle GPS probe and RSD drive data that generalized across cities with no per-city tuning, matching OpenStreetMap and HERE geometry at 95–98% on Dallas and Paris.
- Fine-tuning SAM3 to segment traffic-signal classes in Kanke Gao's group (ongoing); joined the group after an interview following my first three months, and extended the internship to December.
- Developed a geometry-based pipeline to correct position, heading, and road association for 1,790 traffic signs across 5 countries, each fix carrying a calibrated confidence score.
- Trained and benchmarked DETR detectors (RF-DETR with a DINOv2 backbone, RT-DETRv2) for 10-class traffic-sign detection, both exceeding 90% mAP@50.

AI Intern – Samsung Innovation Campus (SIC) Aug 2024 – Nov 2024

- Built end-to-end ML pipelines (preprocessing, model selection, evaluation) and trained CNN, RNN, and GAN models with TensorFlow and Keras for computer vision tasks.

Data Engineering Intern – Digital Egypt Pioneers Initiative (DEPI) Apr 2024 – Oct 2024

- Built data pipelines on Microsoft Azure with SQL-based data warehousing, designed schemas for ML-ready datasets, and set up MLflow and Hugging Face workflows to manage the model lifecycle.

Software Engineer Trainee – ALX Software Engineering Program Jun 2023 – Aug 2024

- Completed 20+ projects across Linux, data structures, full-stack, and systems programming, including a custom *printf* and a custom Unix shell written in C.

AI Mentee – Deloitte Innovation Hub (DIH) Jun 2025 – Aug 2025

- Worked one-on-one with a Deloitte AI engineer on real project workflows, tooling, and industry practice.

Projects

Multi-Modal Road Network Extraction (HERE Hackathon, 1st Place) *PDAL, GeoPandas, LiDAR, Scikit-learn*

- Built a Multi-Modal Semantic Cost-Map that fuses 122M LiDAR points with vehicle probe data; used Road-Aware KDE and RA-Kmeans to rebuild centerlines and classify navigability.
- Won 1st place for geometric precision (6.12 m median error) and physical-constraint robustness (95.5%).

HHTS Superpixel Segmentation *OpenCV, scikit-image*

- Re-implemented a CVPR 2024 method (Hierarchical Histogram Threshold Segmentation) for remote sensing; reached Boundary Recall of **0.865 vs. 0.716 (SLIC)** across 5 aerial images.

Drug–Drug Interaction Prediction (KG-SS-GNN) *PyTorch Geometric, RDKit, GNN*

- Built a Siamese Graph Attention Network that predicts 86 interaction types using BRICS fragmentation, fused with a biological Knowledge Graph and trained with Focal Loss for rare classes.

Other projects

- **ConceptCatch** (LangChain, Azure OpenAI, FastAPI, React): GenAI tool for quiz generation, automatic feedback, and RAG-based document summarization.
- **AI Product Review Summarizer** (BERT, Gemini API, FastAPI): aspect-based sentiment system that extracts product pros and cons.
- **AI-Powered Food Recommendation** (ResNet50, LLaVA): predicts dish names, ingredients, and instructions from food images, trained on 13,582 image–recipe pairs.

Technical Skills

Programming: Python, C, SQL **AI/ML:** Deep Learning, Transformers, CNNs, RNNs, GANs, NLP, Knowledge Distillation, Contrastive Learning
Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Hugging Face, LangChain, GeoPandas, PDAL, Rasterio
Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Knowledge-Graph Grounding, Prompt Engineering
MLOps & Cloud: Kubernetes, MLflow, FastAPI, Microsoft Azure, AWS, Git, GitHub, Linux